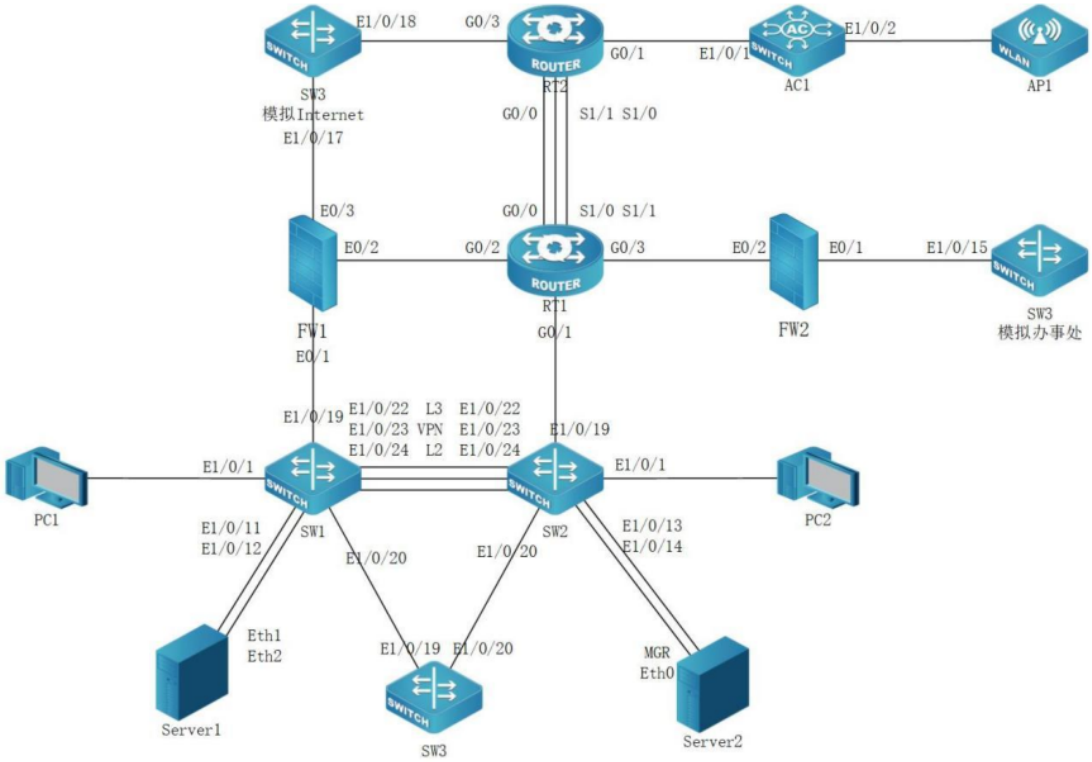
模块二：网络建设与调试

任务描述：

某集团公司原在城市 A 成立了总公司，后在城市 B 成立了分公司，又在城市 C 建立了办事处。集团设有产品、营销、法务、财务、人力 5个部门，统一进行 IP 及业务资源的规划和分配，全网采用 OSPF、RIP、ISIS、BGP 路由协议进行互联互通。随着企业数字化转型工作进一步推进，为持续优化运营创新，充分激活数据要素潜能，为社会创造更多价值，集团决定在总公司建立两个数据中心，在某省建立异地灾备数据中心，以达到快速、可靠交换数据，增强业务部署弹性的目的，完成向两地三中心整体战略架构演进，更好的服务于公司客户。

网络拓扑图及 IP 地址表:

1. 网络拓扑图



设备名称	设备接口	IP 地址
SW1	loopback1 ospfv2 ospfv3 bgp	10.10.1.1/32 2001:10:1:1::1/128
	loopback2	10.10.1.2/32 2001:10:1:1::2/128
	vlan10	10.10.11.1/24 2001:10:1:11::1/64
	vlan20	10.10.12.1/24 2001:10:1:12::1/64
	vlan30	10.10.13.1/24 2001:10:1:13::1/64
	vlan40	10.10.14.1/24 2001:10:1:14::1/64
	vlan50	10.10.15.1/24 2001:10:1:15::1/64
	vlan1019	10.10.255.14/30
	vlan1020	10.10.255.5/30
	vlan1022	10.10.255.1/30
	vlan1023 vpn	10.10.255.1/30

SW2	loopback1 ospfv2 ospfv3 bgp	10.10.10.1/32 2001:10:1:2::1/128
	loopback2	10.10.10.2/32 2001:10:1:2::2/128
	vlan10	10.10.21.1/24 2001:10:1:21::1/64
	vlan20	10.10.22.1/24 2001:10:1:22::1/64
	vlan30	10.10.23.1/24 2001:10:1:23::1/64
	vlan40	10.10.24.1/24 2001:10:1:24::1/64
	vlan50	10.10.25.1/24 2001:10:1:25::1/64
	vlan1019	10.10.255.22/30
	vlan1020	10.10.255.9/30
	vlan1022	10.10.255.2/30
	vlan1023 vpn	10.10.255.2/30

SW3	loopback1 ospfv2 ospfv3 bgp	10.10.3.1/32 2001:10:1:3::1/128
	vlan10	10.10.31.1/24 2001:10:1:31::1/64
	vlan20	10.10.32.1/24 2001:10:1:32::1/64
	vlan30	10.10.33.1/24 2001:10:1:33::1/64
	vlan50	10.10.35.1/24 2001:10:1:35::1/64
	vlan1019	10.10.255.6/30
	vlan1020	10.10.255.10/30
SW3 模拟 办事处	loopback2	10.10.3.2/32 2001:10:1:3::2/128
	vlan110	10.10.110.1/24 2001:10:1:110::1/64
	vlan120	10.10.120.1/24 2001:10:1:120::1/64
	vlan1015	10.10.255.46/30
SW3 模拟 Internet	loopback3	200.200.3.3/32 2001:200:200:3::3/128
	vlan1017	200.200.200.1/30
	vlan1018	200.200.200.5/30

AC1	loopback1 ospfv2 ospfv3	10.10.4.1/32 2001:10:1:4::1/128
	loopback2 rip ripng	10.10.4.2/32 2001:10:1:4::2/128
	loopback3	10.10.4.3/32 2001:10:1:4::3/128
	vlan1001	10.10.255.42/30
	vlan130 无线管理	10.10.130.1/24 2001:10:1:130::1/64
	vlan140 无线 2.4G 产品	10.10.140.1/24 2001:10:1:140::1/64
	vlan150 无线 5G 营销	10.10.150.1/24 2001:10:1:150::1/64

RT1	loopback1 ospfv2 ospfv3 bgp mpls	10.10.5.1/32 2001:10:1:5::1/128
	loopback2 rip ripng	10.10.5.2/32 2001:10:1:5::2/128
	loopback3 isis	10.10.5.3/32 2001:10:1:5::3/128
	loopback4 集团与办事处互联	10.10.5.4/32 2001:10:1:5::4/128
	loopback5 vpn 财务	10.10.5.5/32 2001:10:1:5::5/128
	g0/0	10.10.255.29/30
	g0/1	10.10.255.21/30
	g0/2	10.10.255.18/30
	g0/3	10.10.255.25/30
	s1/0	10.10.255.33/30
	s1/1	10.10.255.37/30

RT2	loopback1 ospfv2 ospfv3 bgp mpls	10.10.6.1/32 2001:10:1:6::1/128
	loopback2 rip ripng	10.10.6.2/32 2001:10:1:6::2/128
	loopback3 isis	10.10.6.3/32 2001:10:1:6::3/128
	loopback4 ipsecvpn	10.10.6.4/32 2001:10:1:6::4/128
	tunnel4 ipsecvpn	10.10.255.50/30
	loopback5 vpn 财务	10.10.6.5/32 2001:10:1:6::5/128
	g0/0	10.10.255.30/30
	g0/1	10.10.255.41/30
	g0/3	200.200.200.6/30
	s1/0	10.10.255.38/30
	s1/1	10.10.255.34/30

FW1	loopback1 ospfv2 ospfv3 trust	10.10.7.1/32 2001:10:1:7::1/128
	loopback2 rip ripng trust	10.10.7.2/32 2001:10:1:7::2/128
	loopback4 ipsecvpn trust	10.10.7.4/32 2001:10:1:7::4/128
	tunnel4 ipsecvpn VPNHUB	10.10.255.49/30
	e0/1 trust	10.10.255.13/30
	e0/2 trust	10.10.255.17/30
	e0/3 untrust	200.200.200.2/30
FW2	loopback1 ospfv2 ospfv3 trust	10.10.8.1/32 2001:10:1:8::1/128
	e0/1 trust	10.10.255.45/30
	e0/2 dmz	10.10.255.26/30

SW1:全局配置模式

```
vlan 10;20;30;40;50;1019-1020;1022-1023
!
interface Vlan10
 ipv6 address 2001:10:1:11::1/64
 ip address 10.10.11.1 255.255.255.0
!
interface Vlan20
 ipv6 address 2001:10:1:12::1/64
 ip address 10.10.12.1 255.255.255.0
!
interface Vlan30
 ipv6 address 2001:10:1:13::1/64
 ip address 10.10.13.1 255.255.255.0
!
interface Vlan40
 ipv6 address 2001:10:1:14::1/64
 ip address 10.10.14.1 255.255.255.0
!
```

```
interface Vlan50
 ipv6 address 2001:10:1:15::1/64
 ip address 10.10.15.1 255.255.255.0
!
interface Vlan1019
 ip address 10.10.255.14 255.255.255.252
!
interface Vlan1020
 ip address 10.10.255.5 255.255.255.252
!
interface Vlan1022
 ip address 10.10.255.1 255.255.255.252
!
interface Loopback1
 ipv6 address 2001:10:1:1::1/128
 ip address 10.10.1.1 255.255.255.255
!
interface Loopback2
 ipv6 address 2001:10:1:1::2/128
 ip address 10.10.1.2 255.255.255.255
!

```

SW2:全局配置模式

```
vlan 10;20;30;40;50;1019-1020;1022-1023
!
interface Vlan10
 ipv6 address 2001:10:1:21::1/64
 ip address 10.10.21.1 255.255.255.0
```

```
!
interface Vlan20
 ipv6 address 2001:10:1:22::1/64
 ip address 10.10.22.1 255.255.255.0
!
interface Vlan30
 ipv6 address 2001:10:1:23::1/64
 ip address 10.10.23.1 255.255.255.0
!
interface Vlan40
 ipv6 address 2001:10:1:24::1/64
 ip address 10.10.24.1 255.255.255.0
!
interface Vlan50
 ipv6 address 2001:10:1:25::1/64
 ip address 10.10.25.1 255.255.255.0
!
interface Vlan1019
 ip address 10.10.255.22 255.255.255.252
!
interface Vlan1020
 ip address 10.10.255.9 255.255.255.252
!
interface Vlan1022
 ip address 10.10.255.2 255.255.255.252
!
interface Loopback1
 ipv6 address 2001:10:1:2::1/128
 ip address 10.10.10.1 255.255.255.255
!
```

```
interface Loopback2
 ipv6 address 2001:10:1:2::2/128
 ip address 10.10.10.2 255.255.255.255
!
```

SW3:全局配置模式

```
vlan 1;10;20;30;50;1015;1017-1020
!
interface Vlan10
 ipv6 address 2001:10:1:31::1/64
 ip address 10.10.31.1 255.255.255.0
!
interface Vlan20
 ipv6 address 2001:10:1:32::1/64
 ip address 10.10.32.1 255.255.255.0
!
interface Vlan30
 ipv6 address 2001:10:1:33::1/64
 ip address 10.10.33.1 255.255.255.0
!
interface Vlan50
 ipv6 address 2001:10:1:35::1/64
 ip address 10.10.35.1 255.255.255.0
!
interface Vlan1019
 ip address 10.10.255.6 255.255.255.252
!
interface Vlan1020
 ip address 10.10.255.10 255.255.255.252
```

```
!
interface Loopback1
 ipv6 address 2001:10:1:3::1/128
 ip address 10.10.3.1 255.255.255.255
!
```

AC1:全局配置模式

```
vlan 130;140;150;1001
!
interface Vlan130
 ipv6 address 2001:10:1:130::1/64
 ip address 10.10.130.1 255.255.255.0
!
interface Vlan140
 ipv6 address 2001:10:1:140::1/64
 ip address 10.10.140.1 255.255.255.0
!
interface Vlan150
 ipv6 address 2001:10:1:150::1/64
 ip address 10.10.150.1 255.255.255.0
!
interface Vlan1001
 ip address 10.10.255.42 255.255.255.252
!
interface Loopback1
 ipv6 address 2001:10:1:4::1/128
 ip address 10.10.4.1 255.255.255.255
!
interface Loopback2
```

```
 ipv6 address 2001:10:1:4::2/128
 ip address 10.10.4.2 255.255.255.255
!
interface Loopback3
 ipv6 address 2001:10:1:4::3/128
 ip address 10.10.4.3 255.255.255.255
!
```

RT1:全局配置模式

```
interface Loopback1
 ip address 10.10.5.1 255.255.255.255
 ipv6 address 2001:10:1:5::1/128
!
interface Loopback2
 ip address 10.10.5.2 255.255.255.255
 ipv6 address 2001:10:1:5::2/128
!
interface Loopback3
 ip address 10.10.5.3 255.255.255.255
 ipv6 address 2001:10:1:5::3/128
!
interface Loopback4
 ip address 10.10.5.4 255.255.255.255
 ipv6 address 2001:10:1:5::4/128
!
interface Loopback5
 ip address 10.10.5.5 255.255.255.255
 ipv6 address 2001:10:1:5::5/128
!
```

```
interface GigaEthernet0/0
 ip address 10.10.255.29 255.255.255.252
ipv6 enable
!
interface GigaEthernet0/1
 ip address 10.10.255.21 255.255.255.252
ipv6 enable
!
interface GigaEthernet0/2
 ip address 10.10.255.18 255.255.255.252
ipv6 enable
!
interface GigaEthernet0/3
 ip address 10.10.255.25 255.255.255.252
 ipv6 enable
!
interface Serial1/0
 ip address 10.10.255.33 255.255.255.252
 physical-layer speed 2048000
ipv6 enable
!
interface Serial1/1
 ip address 10.10.255.37 255.255.255.252
ipv6 enable
!
```

RT2:全局配置模式

```
interface Loopback1
 ip address 10.10.6.1 255.255.255.255
```

```
 ipv6 address 2001:10:1:6::1/128
!
interface Loopback2
 ip address 10.10.6.2 255.255.255.255
 ipv6 address 2001:10:1:6::2/128
!
interface Loopback3
 ip address 10.10.6.3 255.255.255.255
 ipv6 address 2001:10:1:6::3/128
!
interface Loopback4
 ip address 10.10.6.4 255.255.255.255
 ipv6 address 2001:10:1:6::4/128
!
interface Loopback5
 ip address 10.10.6.5 255.255.255.255
 ipv6 address 2001:10:1:6::5/128
!
interface Tunnel4
 ip address 10.10.255.50 255.255.255.252
 tunnel speed-up
!
interface GigaEthernet0/0
 ip address 10.10.255.30 255.255.255.252
ipv6 enable
!
interface GigaEthernet0/1
 ip address 10.10.255.41 255.255.255.252
ipv6 enable
!
```

```
interface GigaEthernet0/2
 no ip address
!
interface GigaEthernet0/3
 ip address 200.200.200.6 255.255.255.252
!
interface Serial1/0
 ip address 10.10.255.38 255.255.255.252
 physical-layer speed 2048000
ipv6 enable
!
interface Serial1/1
 ip address 10.10.255.34 255.255.255.252
ipv6 enable
!
```

FW1:全局配置模式

```
interface ethernet0/1
  zone  "trust"
  ip address 10.10.255.13 255.255.255.252
  ipv6 enable
manage ping
  manage ssh
  manage http
y
  manage https
exit
```

```
interface ethernet0/2
  zone  "trust"
  ip address 10.10.255.17 255.255.255.252
  ipv6 enable
manage ping
  manage ssh
  manage http
y
  manage https
exit
interface ethernet0/3
  zone  "untrust"
  ip address 200.200.200.2 255.255.255.252
manage ping
  manage ssh
  manage http
y
  manage https
exit
interface loopback1
  zone  "trust"
  ip address 10.10.7.1 255.255.255.255
  ipv6 enable
  ipv6 address 2001:10:1:7::1/128
  manage ping
  manage ssh
  manage http
y
```

```
  manage https
exit
interface loopback2
  zone  "trust"
  ip address 10.10.7.2 255.255.255.255
  ipv6 enable
  ipv6 address 2001:10:1:7::2/128
  manage ping
  manage ssh
  manage http
y
  manage https
exit
interface loopback4
  zone  "trust"
  ip address 10.10.7.4 255.255.255.255
  ipv6 enable
  ipv6 address 2001:10:1:7::4/128
  manage ping
  manage ssh
  manage http
y
  manage https
exit
interface tunnel4
  zone  "VPNHub"
  ip address 10.10.255.49 255.255.255.252
  manage ping
  manage ssh
```

```
  manage http
y
  manage https
exit
```

FW2:全局配置模式

```
interface ethernet0/1
  zone  "trust"
  ip address 10.10.255.45 255.255.255.252
  ipv6 enable
manage ping
  manage ssh
  manage http
y
  manage https
exit
interface ethernet0/2
  zone  "dmz"
  ip address 10.10.255.26 255.255.255.252
  ipv6 enable
manage ping
  manage ssh
  manage http
y
  manage https
exit
interface loopback1
```

```
  zone  "trust"
  ip address 10.10.8.1 255.255.255.255
  ipv6 enable
  ipv6 address 2001:10:1:8::1/128
  manage ping
  manage ssh
  manage http
y
  manage https
exit
```